TOPOLOGIES

Course Name	Computer Networks	Course Code	CSA0709
Student Name	AB SAI SARANYA	Register No.	1921260020

i. Bus Topology

Procedure

- Take 5 switches (Switch-PT) and place them in a line.
- Connect one PC to each switch, placed opposite to one another.
- Enter the IP configuration for all the PCs.
- Send a message from one PC to another, run the simulation and get the output as successful.

Output (Cisco Packet Tracer)

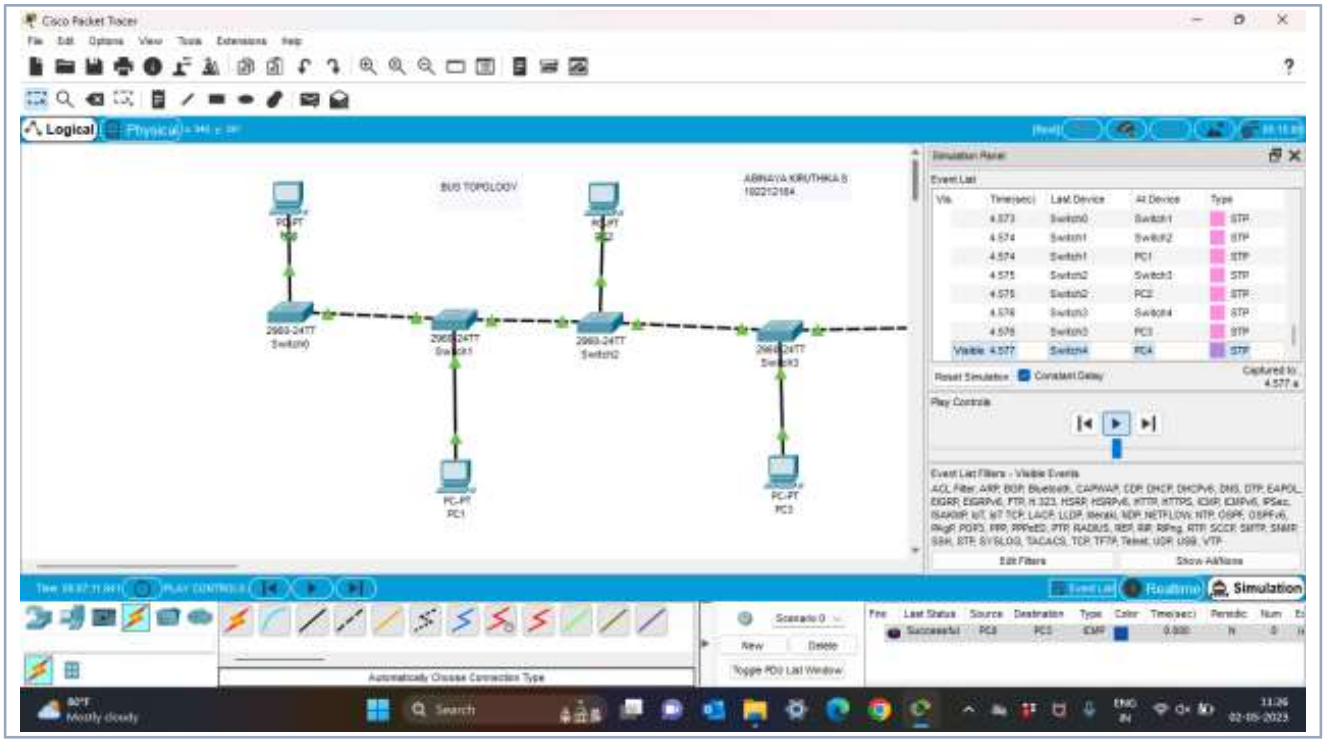


Fig. 1 – Bus topology simulation in Cisco Packet Tracer

Result

The bus topology was created in Cisco Packet Tracer, the message was sent from one device to another in simulation mode, and the output was **successful**.

ii. Star Topology

Procedure

- Take 4 PCs and 1 Hub (Hub-PT) and place the hub in the centre.
- Connect all the PCs to the hub.
- Enter the IP configuration for all 4 PCs.
- Send a message, start the simulation and get the output as successful.

Output (Cisco Packet Tracer)

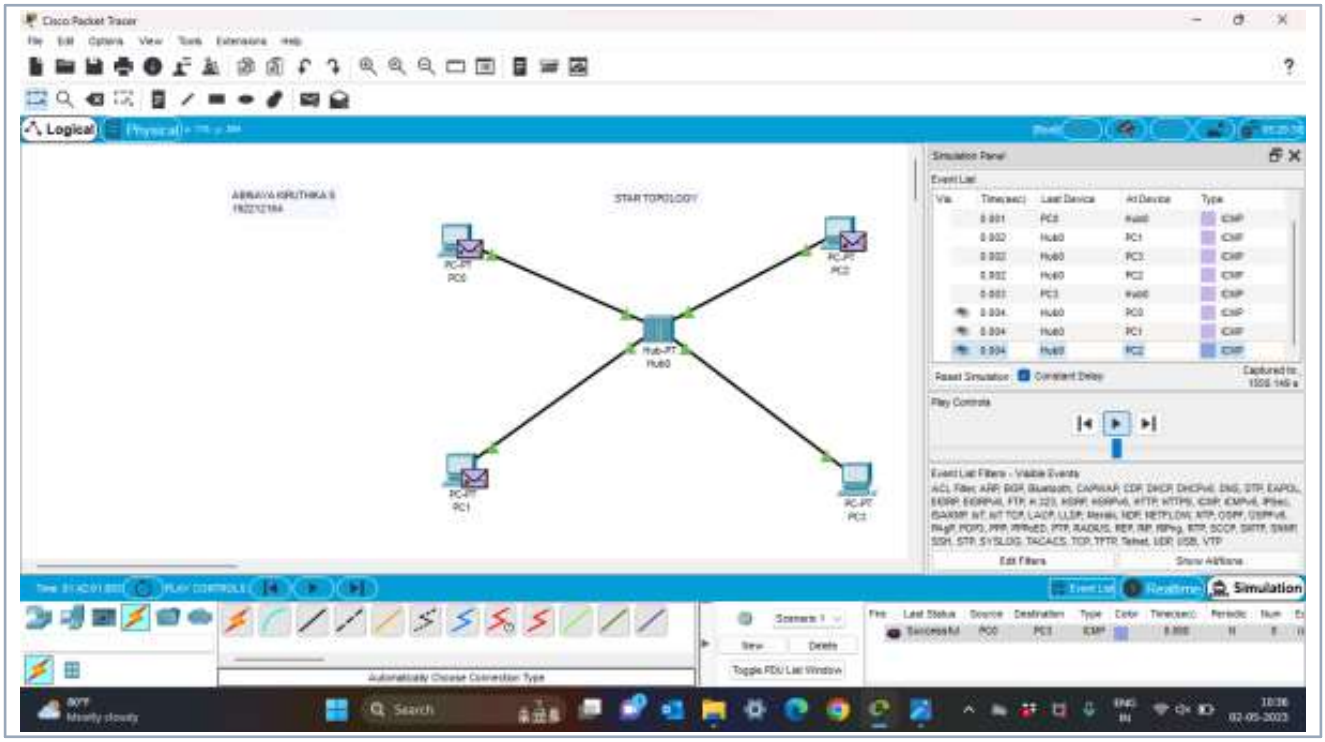


Fig. 2 – Star topology simulation in Cisco Packet Tracer

Result

The star topology was created in Cisco Packet Tracer, the message was sent from one device to another in simulation mode, and the output was **successful**.

iii. Ring Topology

Procedure

- Take 4 switches (Switch-PT) and 4 laptops (Laptop-PT).
- Connect the switches in a closed loop and attach one laptop to each switch.
- Enter the IP configuration for all 4 laptops.
- Send a message, start the simulation and get the output as successful.

Output (Cisco Packet Tracer)

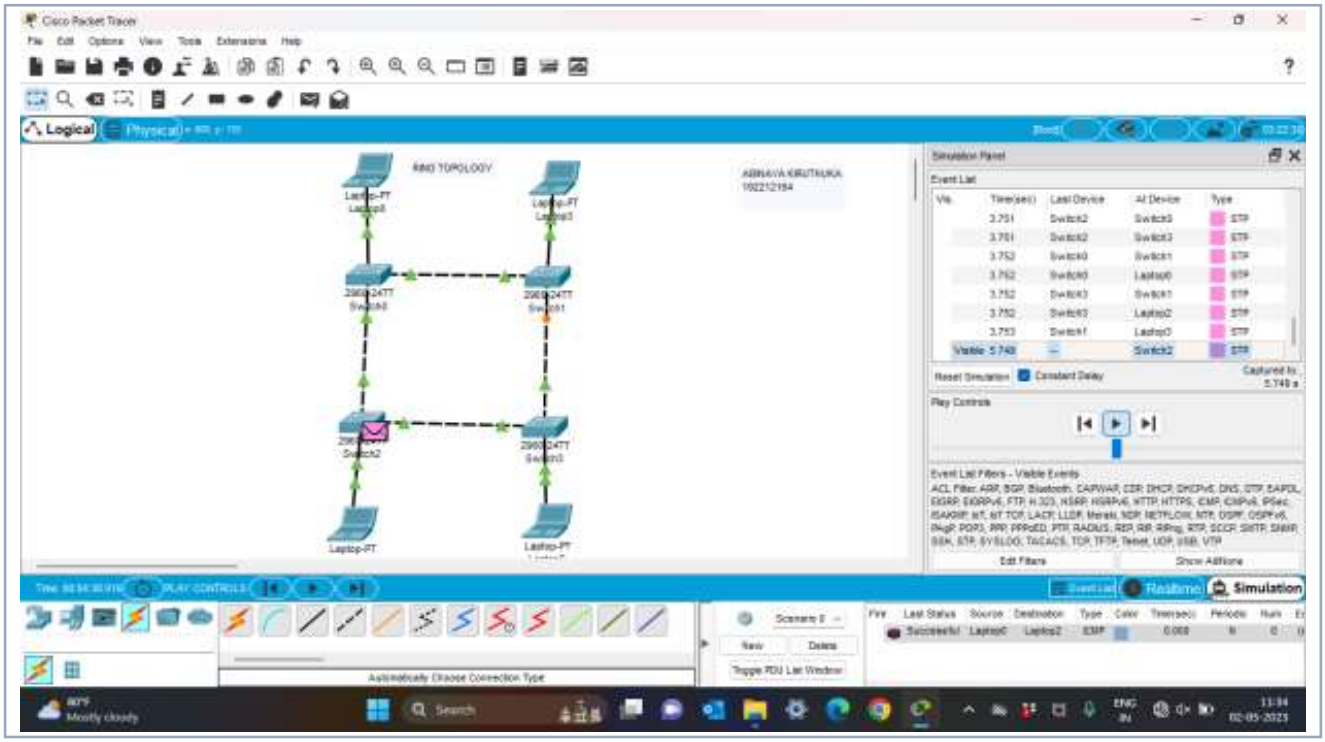


Fig. 3 – Ring topology simulation in Cisco Packet Tracer

Result

The ring topology was created in Cisco Packet Tracer, the message was sent from one device to another in simulation mode, and the output was **successful**.

iv. Mesh Topology

Procedure

- Take 5 switches (Switch-PT) and 5 PCs (PC-PT).
- Connect every switch to every other switch, then attach one PC to each switch.
- Enter the IP configuration for all the PCs.
- Send a message from one PC to another, start the simulation and get the output as successful.

Output (Cisco Packet Tracer)

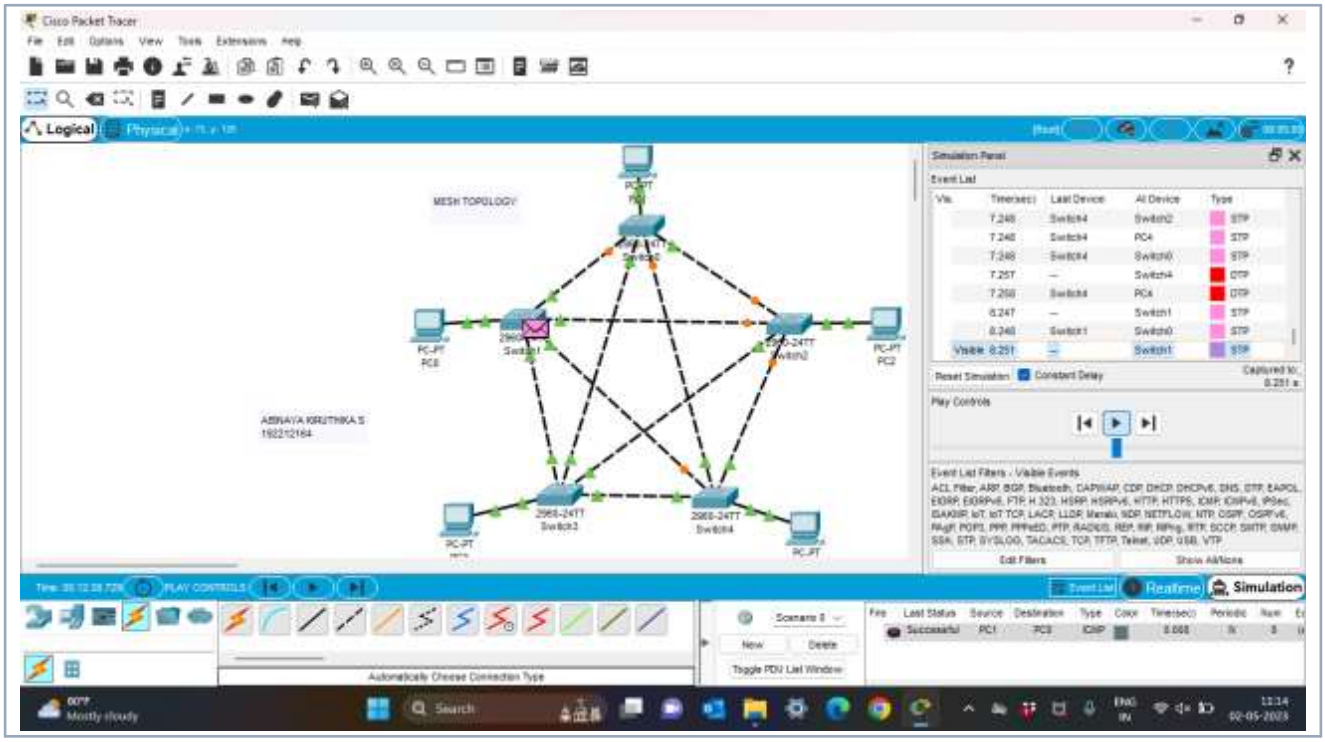


Fig. 4 – Mesh topology simulation in Cisco Packet Tracer

Result

The mesh topology was created in Cisco Packet Tracer, the message was sent from one device to another in simulation mode, and the output was **successful**.

v. Tree Topology

Procedure

- Take 8 switches (Switch-PT) and 8 laptops, and connect one laptop to each switch.
- Connect the switches in a tree (hierarchical) layout.
- Enter the IP configuration for all the laptops.
- Send messages from one laptop to another, start the simulation and get the output as successful.

Output (Cisco Packet Tracer)

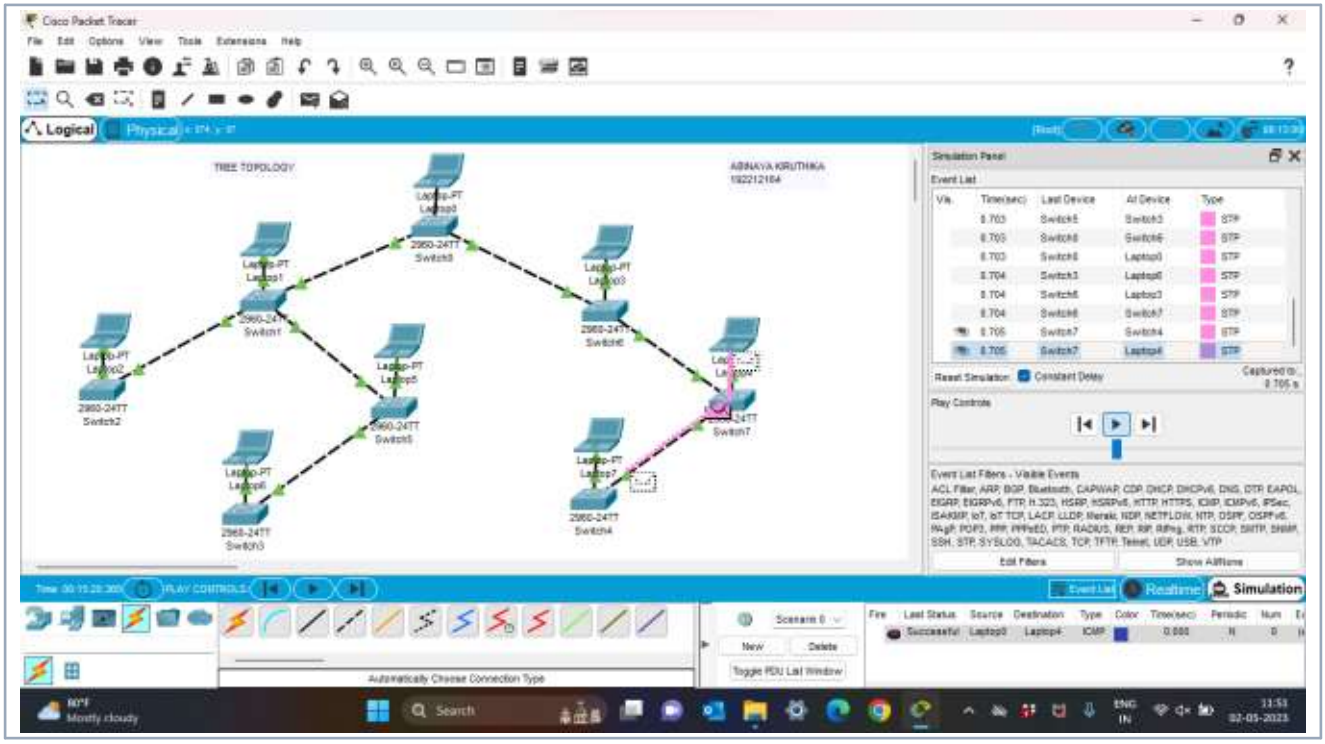


Fig. 5 – Tree topology simulation in Cisco Packet Tracer

Result

The tree topology was created in Cisco Packet Tracer, the message was sent from one device to another in simulation mode, and the output was **successful**.